Selection of Baseline Framework

What it is. kvpress is a mature, actively maintained framework (Apache-2.0) for **KV-cache compression**: it plugs cache-eviction policies into HuggingFace `transformers` via a forward hook. The central abstraction is the `ScorerPress`: a subclass implements `score(...) -> (batch, num_kv_heads, seq_len)` and the framework keeps the top-scoring tokens during prefill. It ships ~40 published presses (KnormPress, SnapKVPress, ExpectedAttentionPress, the leverage-score family, …).

Default / baseline policy. Independent, per-token scoring: each press ranks tokens on its own signal (key norm, recent-query attention, expected attention, etc.) and the framework evicts the lowest-scoring `compression_ratio` fraction. The natural baseline for this project is `KnormPress` (keep low-L2-norm keys), keys-only, essentially free, and the framework's smooth default scorer.

Why chosen. It is the de-facto standard harness for this task: a shared evaluation stack (RULER, LongBench, a public leaderboard, a tiny CPU unit-test model), so a new press can be benchmarked against many strong baselines under **identical workloads** with minimal glue. Its `ScorerPress` contract makes a new relational scorer a ~150-line drop-in, and its PR history merges external and even AI-authored scorers, so the contribution has a real upstream path.

What this project adds. Existing scorers rank tokens *independently* and ignore the relational structure of the cache (a token can be individually unremarkable yet crucial because it *supports* an important one). `CentralityPress` scores tokens by centrality in the key-similarity graph, anchored to `KnormPress` via a personalized-PageRank teleport, keeping needles in and repairing the base's collapse under compression.